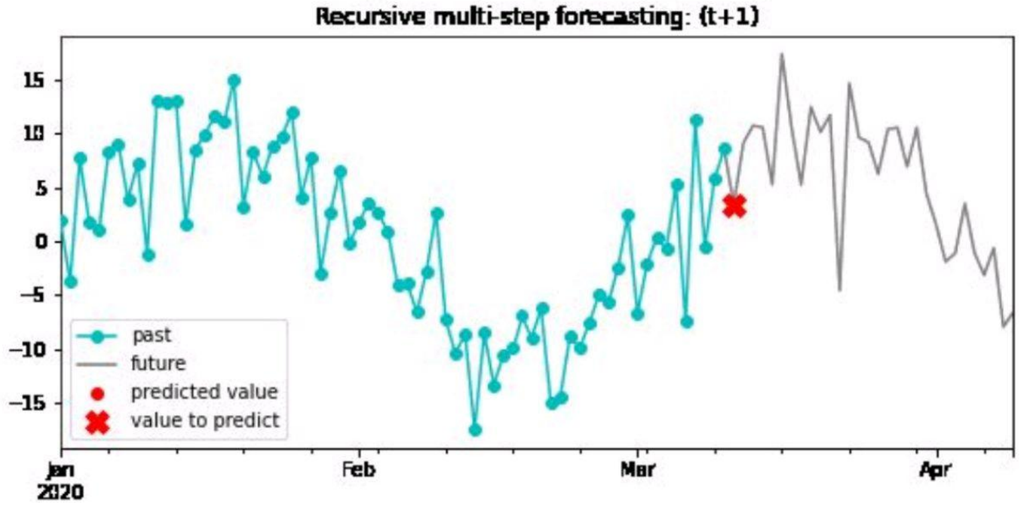
Time series forecasting



Time series data

date	charter_foreign	total_foreign	charter_domestic	total_domestic
date	bigint	bigint	bigint	bigint
Date	integer	integer	integer	integer
2008-01-01T00:00:00.000Z	57463	5642057	174692	7394181
2008-02-01T00:00:00.000Z	56651	5053435	142931	6837201
2008-03-01T00:00:00.000Z	95878	6071551	176906	8439191
2008-04-01T00:00:00.000Z	100198	5731709	103054	7467710
2008-05-01T00:00:00.000Z	130720	6003335	88631	7840260
2008-06-01T00:00:00.000Z	196656	6467543	99579	8274573
2008-07-01T00:00:00.000Z	195873	6861571	130766	8971734
2008-08-01T00:00:00.000Z	152848	6895232	77253	8717085
2008-09-01T00:00:00.000Z	88267	5539617	40070	6371669
2008-10-01T00:00:00.000Z	76723	5564387	43003	6520788
2008-11-01T00:00:00.000Z	26671	5128350	50322	6223857
2008-12-01T00:00:00.000Z	30963	5624131	54894	6959043
2009-01-01T00:00:00.000Z	35174	5399479	60652	6805171

	Jan-17	Feb-17	Mar-17	Apr-17	May-17	Jun-17	Jul-17	Aug-17	Sep-17	Oct-17	Nov-17	Dec-17
Store 1	20 809	23 832	22 542	18 736	15 347	12 190	11 049	10 913	9 019	9 803	3 172	7 768
Store 2	14 834	14 170	13 109	7 560	6 388	8 124	6 719	3 847	3 187	8 145	10 007	10 036
Store 3	14 453	12 816	14 260	12 954	12 596	11 200	8 374	9 725	10 853	11 609	13 897	19 628
Store 4	8 590	8 822	13 497	13 073	13 278	14 705	13 550	13 962	14 783	11 712	13 234	17 790
Store 5	16 647	12 851	13 433	13 486	13 767	10 482	11 476	13 024	14 459	14 672	12 885	16 681
Store 6	7 323	5 043	4 857	2 584	6 990	4 130	45 110	4 895	5 708	5 106	4 334	10 741
Store 7	8 735	4 645	4 905	4 536	5 385	5 040	6 341	9 403	9 904	9 521	13 473	7 077
Store 8	965	1 254	1 937	1 940	1 756	1 690	1 536	2 502	2 806	4 087	3 704	5 478
Store 9	188	4 375	5 109	5 255	5 667	6 331	9 182	9 867	8 948	10 096	10 860	10 951
Store 10	14 033	30 900	18 864	12 828	10 269	8 464	7 674	4 160	5 497	9 608	16 177	15 798
Store 11	8 185	9 090	12 560	13 682	10 061	8 172	8 045	6 622	7 240	9 923	9 403	10 768
Store 12	911	384	658	949	1 236	1 405	1 366	1 433	1 282	1 515	1 457	2 933
Store 13	2 361	2 830	2 892	2 953	2 929	2 982	6 361	25 969	22 952	27 171	27 199	35 298
Store 14	9 805	8 436	12 878	9 580	6 665	5 069	6 003	6 139	10 987	11 039	11 252	13 844

Code	Name	Population|1990	Population|2000	Population|2010	Population Change|1980-1990	Population Change|1990-2000
AK	Alaska	550043	626932	710231	36.9	14
AL	Alabama	4040587	4447100	4779736	3.8	10.1
AR	Arkansas	2350725	2673400	2915918	2.8	13.7
AZ	Arizona	3665228	5130632	6392017	34.8	40
CA	California	29760021	33871648	37253956	25.7	13.8
CO	Colorado	3294394	4301261	5029196	14	30.6
CT	Connecticut	3287116	3405565	3574097	5.8	3.6
DC	District of Columbia	606900	572059	601723	-4.9	-5.7
DE	Delaware	666168	783600	897934	12.1	17.6
FL	Florida	12937926	15982378	18801310	32.7	23.5
GA	Georgia	6478216	8186453	9687653	18.6	26.4
HI	Hawaii	1108229	1211537	1360301	14.9	9.3
IA	Iowa	2776755	2926324	3046355	-4.7	5.4
ID	Idaho	1006749	1293953	1567582	6.7	28.5
IL	Illinois	11430602	12419293	12830632	0	8.6
IN	Indiana	5544159	6080485	6483802	1	9.7
KS	Kansas	2477574	2688418	2853118	4.8	8.5
KY	Kentucky	3685296	4041769	4339367	0.7	9.7
LA	Louisiana	4219973	4468976	4533372	0.3	5.9

Time series data

1	2	3	4	5	6	7	8	9	10

Time series

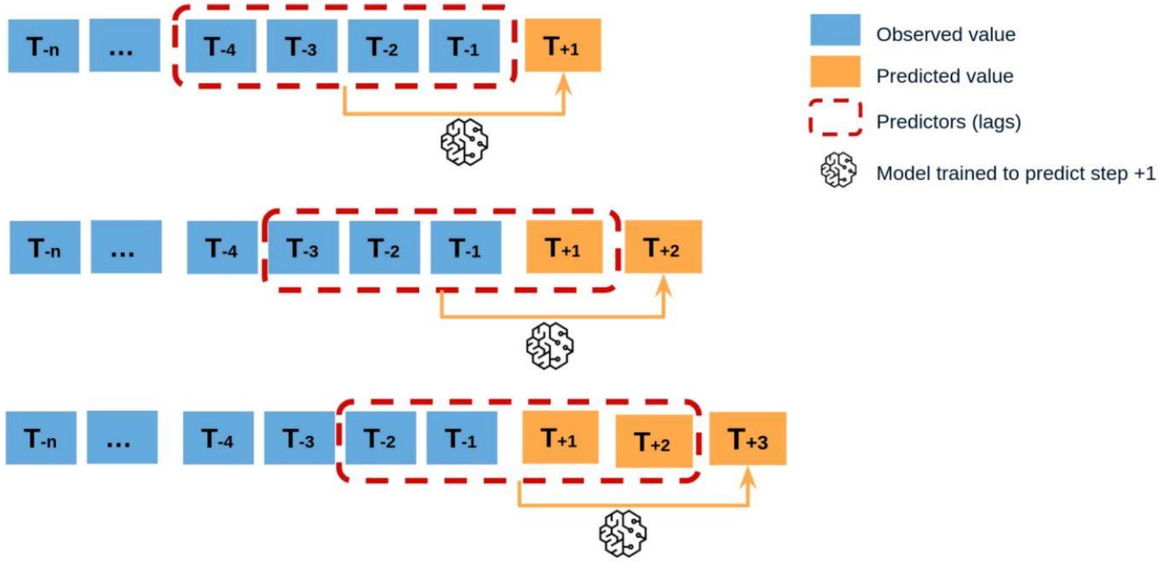
1	2	3	4	5	6	7	8	9	10

Exogenous variable

a	b	c	d	e	f	g	h	i	j

X						y
1	2	3	4	5	f	6
2	3	4	5	6	g	7
3	4	5	6	7	h	8
4	5	6	7	8	i	9
5	6	7	8	9	j	10

Recursive multi-step Time series Forecasting



Direct multi-step Time series Forecasting

